

Lei Chen

Ph.D.

PROFILE

Lei Chen is a Postdoctoral Fellow at the **Zhejiang University-University of Edinburgh Institute, International Campus, Zhejiang University**. He received his **Ph.D. in Biochemistry and Molecular Biology (Biomaterials)** from **East China Normal University**. His current research focuses on biomaterials, wound healing and stem cell-based therapy for tissue regeneration.

EDUCATION

- East China Normal University** Sep. 2019 – Jun. 2025
Ph.D. in Biochemistry and Molecular Biology (Biomaterials) Shanghai, China
 - GPA: 3.60/4.00
 - Supervisor: **Prof. Qiang Zhang**
- Zhejiang A&F University** Sep. 2015 – Jun. 2019
B.Sc. in Applied Chemistry Hangzhou, China
 - GPA: 88.6/100 (Ranked 1st in major)
 - Awarded provincial and university-level scholarships for academic excellence

PROJECTS

- Photosynthetic Microalgae–MSC Symbiotic Hydrogel for Chronic Non-Healing Wounds** 2025 – Present
Postdoctoral Project, Zhejiang University-University of Edinburgh Institute
 - Investigating a **photosynthetic microalgae–MSC symbiotic system** to actively remodel the pathological microenvironment of chronic non-healing wounds.
 - Developing an **in situ-gelling biohybrid hydrogel** via mild enzymatic crosslinking to co-encapsulate microalgae and mesenchymal stem cells for sustained light-triggered oxygen generation.
 - Studying how photosynthetic oxygenation regulates **MSC energy metabolism, HIF-1 α signaling, and extracellular vesicle secretion** under persistent hypoxia.
 - Elucidating how **photosynthetic potentiation and dual-source bioactive component synergy** promote angiogenesis, inflammation resolution, and tissue reconstruction.
 - Evaluating therapeutic efficacy in multiple **preclinical wound models** to establish a new strategy for active modulation of the hypoxic wound microenvironment.
- Superhydrophobic Nanocellulose Aerogel for Hemostatic Applications** 2020 – 2023
Material Synthesis & Characterization
 - Developed a novel **bio-aerogel** from bacterial cellulose nanofibers with superhydrophobic properties.
 - Achieved **rapid clot formation** in *in vitro* and *in vivo* tests with minimal side effects.
 - Utilized **SEM, TEM, and DMA** to characterize morphological and mechanical properties.
- Inorganic-Organic Composite for Advanced Hemostasis** 2021 – 2024
Zeolite & Nanozeolites In Situ Growth & Nanosilica
 - Engineered **hybrid aerogels** with nanocellulose and inorganic components for enhanced hemostatic coagulation.
 - Tested performance in **femoral bleeding** and **liver laceration** models.
 - Showed superior **clotting efficacy** and **biosafety** profile compared to commercial products.
- Biomimetic Biomass-Based Hemostatic Materials** 2023 – 2025
Advanced Hemostats & Hemostatic Mechanism Investigation
 - Developed biomass-derived materials that **mimic the hemostatic mechanism** of zeolites.
 - Engineered **porous structures** to enhance fluid absorption and promote concentration of clotting factors.
 - Evaluated **biocompatibility and hemostatic efficiency** in *in vitro* and *in vivo* models, demonstrating effective bleeding control.

SKILLS

- Laboratory Techniques:** SEM, TEM, Dynamic Light Scattering (DLS), Rheometry, X-ray Diffraction (XRD), Universal Testing/DMA, Laser Confocal Scanning Microscopy (CLSM), Cell Culture, Animal Bleeding Model
- Software:** Adobe Illustrator, Origin, GraphPad Prism, ImageJ, Blender
- Languages:** Japanese (CEFR A1), English (CET-4: 614, CET-6: 576), Mandarin Chinese (Native)
- Certifications:** High School Chemistry Teaching Certificate, CATTI (Translation) Level 3, LSCAT s220

PUBLICATIONS

1. **Chen, L.**; Xia, R.; Jin, Z.; Li, W.; Wang, G.; Zhang, Q. Calcium-Releasing Silica-Coated Cellulose Nanofiber Aerogel for Safe and Efficient Hemostasis. **Biomater. Adv.** 2026, 215061. <https://doi.org/10.1016/j.bioadv.2026.215061>.
2. Wang, J.; Chang, Y.; Liu, X.; **Chen, L.**; Tian, Y.; Huang, X.; Zhang, Q. A Highly Ductile and Multifunctional Cellulose/CaCl₂ Composite Film for Humidity-Responsive Wearable Sensors. **Int. J. Biol. Macromol.** 2026, 358, 151743. <https://doi.org/10.1016/j.ijbiomac.2026.151743>.
3. Liu, X.; Tian, Y.; Kang, S.; Jin, Z.; **Chen, L.**; Lu, Q.; Wang, L.; Zhang, Q. Mechanically Robust, Forgeable, and Machinable Cellulose Hydrogel. **ACS Sustainable Chem. Eng.** 2026. <https://doi.org/10.1021/acssuschemeng.5c13794>.
4. Xia, R.; Wang, L.; **Chen, L.**; Liu, X.; Chang, Y.; Tian, Y.; Jin, Z.; Zhang, Q. Facile Synthesis of Regenerated Chitosan Nanowhiskers with Concentration-Dependent Self-Assembly Behavior. **Carbohydr. Polym.** 2026, 124615. <https://doi.org/10.1016/j.carbpol.2025.124615>.
5. Chang, Y.; Tian, Y.; Wang, J.; Zhao, J.; **Chen, L.**; Kang, S.; Lu, Q.; He, X.; Zhang, Q. High-Elongation, Water-Weldable, and Fully Degradable Biomass Foams Fabricated via Oven Drying. **Sci. Adv.** 2025, 11 (31), eady0746. <https://doi.org/10.1126/sciadv.ady0746>.
6. **Chen, L.**; Jin, Z.; Kamiya, T.; Liu, X.; Tian, Y.; Xia, R.; Wang, G.; Li, T.; Zhang, Q. Nanoporous Zeolite Anchored Cellulose Nanofiber Aerogel for Safe and Efficient Hemostasis. **Small** 2025, 21 (21), 2500696. <https://doi.org/10.1002/smll.202500696>.
7. Tian, Y.; **Chen, L.**; Liu, X.; Chang, Y.; Xia, R.; Zhang, J.; Kong, Y.; Gong, Y.; Li, T.; Wang, G.; Zhang, Q. Colored Cellulose Nanoparticles with High Stability and Easily Modified Surface for Accurate and Sensitive Multiplex Lateral Flow Assay. **ACS Nano** 2025, 19 (4), 4704–4717. <https://doi.org/10.1021/acsnano.4c15340>.
8. Wang, L.; Tian, Y.; Chang, Y.; **Chen, L.**; Zhang, Q. Cellulose Nanofiber-Created Air Barrier Enabling Closed-Cell Foams Prepared via Oven-Drying. **Carbohydr. Polym.** 2025, 351, 123096. <https://doi.org/10.1016/j.carbpol.2024.123096>.
9. **Chen, L.**; Pan, S.; Wang, L.; Liu, X.; Jin, Z.; Xia, R.; Chang, Y.; Tian, Y.; Gong, Y.; Wang, G.; Zhang, Q. Superhydrophobic Cellulose Nanofiber Aerogels for Efficient Hemostasis with Minimal Blood Loss. **ACS Appl. Mater. Interfaces** 2024, 16 (36), 47294–47302. <https://doi.org/10.1021/acsaami.4c10505>.
10. Jin, Z.; **Chen, L.**; Liu, X.; Xia, R.; Li, W.; Wang, G.; Zhang, Q. Zeolite Firmly Anchored Regenerated Cellulose Aerogel for Efficient and Biosafe Hemostasis. **Int. J. Biol. Macromol.** 2025, 304, 140743. <https://doi.org/10.1016/j.ijbiomac.2025.140743>.
11. Feng, Y.; Chang, Y.; Wang, L.; Liu, X.; **Chen, L.**; Yan, X.; Zhang, Q. Ultrahigh-Strength Cellulose Nanofiber Foams via the Synergy of Freeze-Casting and Solvent Exchange. **Carbohydr. Polym.** 2025, 347, 122671. <https://doi.org/10.1016/j.carbpol.2024.122671>.
12. Tian, Y.; Kong, Y.; Liu, X.; **Chen, L.**; Wang, L.; Zhou, L.; Wang, G.; Zhang, Q. Regenerated Carboxymethyl Cellulose Beads for Selective Removal of Low-Density Lipoprotein from Whole Blood. **Carbohydr. Polym.** 2025, 348, 122848. <https://doi.org/10.1016/j.carbpol.2024.122848>.
13. Liu, X.; Tian, Y.; Wang, L.; **Chen, L.**; Jin, Z.; Zhang, Q. A Cost-Effective and Chemical-Recycling Approach for Facile Preparation of Regenerated Cellulose Materials. **Nano Lett.** 2024, 24 (29), 9074–9081. <https://doi.org/10.1021/acs.nanolett.4c02351>.
14. Pan, S.; Li, Y.; Tong, X.; **Chen, L.**; Wang, L.; Li, T.; Zhang, Q. Strongly-Adhesive Easily-Detachable Carboxymethyl Cellulose Aerogel for Noncompressible Hemorrhage Control. **Carbohydr. Polym.** 2023, 301, 120324. <https://doi.org/10.1016/j.carbpol.2022.120324>.

PATENTS

1. Zhang, Q.; **Chen, L.**; Jin, Z.; Tian, Y. Preparation Method of In Situ Grown Nano-Zeolite Nanocellulose Aerogel and Its Hemostatic Application. **Patent** 2025, Patent No. 202510141274.0.
2. Zhang, Q.; Jin, Z.; **Chen, L.** Preparation of Calcium-Ion-Bound Carboxymethyl Nanocellulose Gauze and Its Hemostatic Application. **Patent** 2025, Patent No. 202510497801.1.
3. Zhang, Q.; Tian, Y.; Li, T.; **Chen, L.** Colored Cellulose Nanoparticles and Their Preparation Method and Application. **Patent** 2024, Patent No. 202410928182.2.
4. Zhang, Q.; Tian, Y.; Wang, L.; Chang, Y.; **Chen, L.** Cellulose Foam Materials and Their Preparation Method and Application. **Patent** 2024, Patent No. 202410928190.7.

LEADERSHIP EXPERIENCE

- **Lab Manager**
School of Life Sciences, East China Normal University
 - Managed lab inventory, procurement, and vendor negotiation.
 - Supervised and trained undergraduate assistants on safety protocols, equipment operation, and data workflows.
 - Implemented a digital scheduling system for instrument bookings, increasing equipment utilization efficiency.
 - Coordinated cross-laboratory collaborations.
 - **Chair, Graduate Student Union**
School of Life Sciences, East China Normal University
 - Oversaw academic seminars, community outreach, and social events for over 200 graduate students.
 - Enhanced departmental engagement by introducing new student-led symposiums.
 - **Director, New Media Center**
School of Sciences, Zhejiang A&F University
 - Managed digital platforms and university-wide communications.
 - Organized promotional campaigns for academic conferences and cultural festivals.

2021 – 2024

2020 – 2021

2016 – 2017

SELECTED HONORS AND AWARDS

- **Postdoctoral Fellowship Program (Grade C)**
China Postdoctoral Science Foundation
 - **Distinguished Graduate Student of Shanghai**
Shanghai Municipal Education Commission
 - Recognized as one of the top graduates in Shanghai for outstanding academic and extracurricular achievements.
 - **National Scholarship (Ph.D. Level)**
Ministry of Education, China
 - Awarded for exceptional academic performance and research excellence during doctoral studies.
 - **National First Prize, 19th Challenge Cup Special Contest**
Central Committee of the Communist Youth League, China Association for Science and Technology
 - Participated in a team project on advanced nanocellulose-based materials.
 - **Outstanding Graduate Student Union Leader**
East China Normal University
 - Honored for leadership in organizing academic and cultural events.
 - **Outstanding Student**
East China Normal University
 - Recognized for exemplary academic performance and campus involvement.
 - **Provincial Second Prize, 15th Challenge Cup National College Student Academic Works Competition**
Zhejiang Provincial Committee for College Student Science and Technology Competition
 - Awarded for an innovative extracurricular research project.

2026

2025

2024

2024

2020

2019

2017

PROFESSIONAL MEMBERSHIPS

- **Chinese Society for Biomaterials (CSBM)**, Member ID: U55OBT7c
 - **Chinese Chemical Society (CCS)**, Member ID: 220511220

2024 – Present

2022 – Present

REFERENCES

1. **Prof. Min Zhou**
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Postdoctoral Supervisor
 2. **Prof. Qiang Zhang**
School of Life Sciences, East China Normal University
qzhang@bio.ecnu.edu.cn
Ph.D. Supervisor
 3. **Prof. Linfang Shi**
College of Chemistry and Materials Engineering, Zhejiang A&F University
lfshi2003@163.com
Undergraduate Tutor

ADDITIONAL INFORMATION

Date of Birth: August 5, 1997
Current Position: Postdoctoral Fellow, Zhejiang University-University of Edinburgh Institute, International Campus, Zhejiang University